INF1002
Programming Fundamentals
Lecture 2: Python basic II

Review

- Variables
  - Id
  - Name
  - Type and Type Casting
  - Arithmetic operators

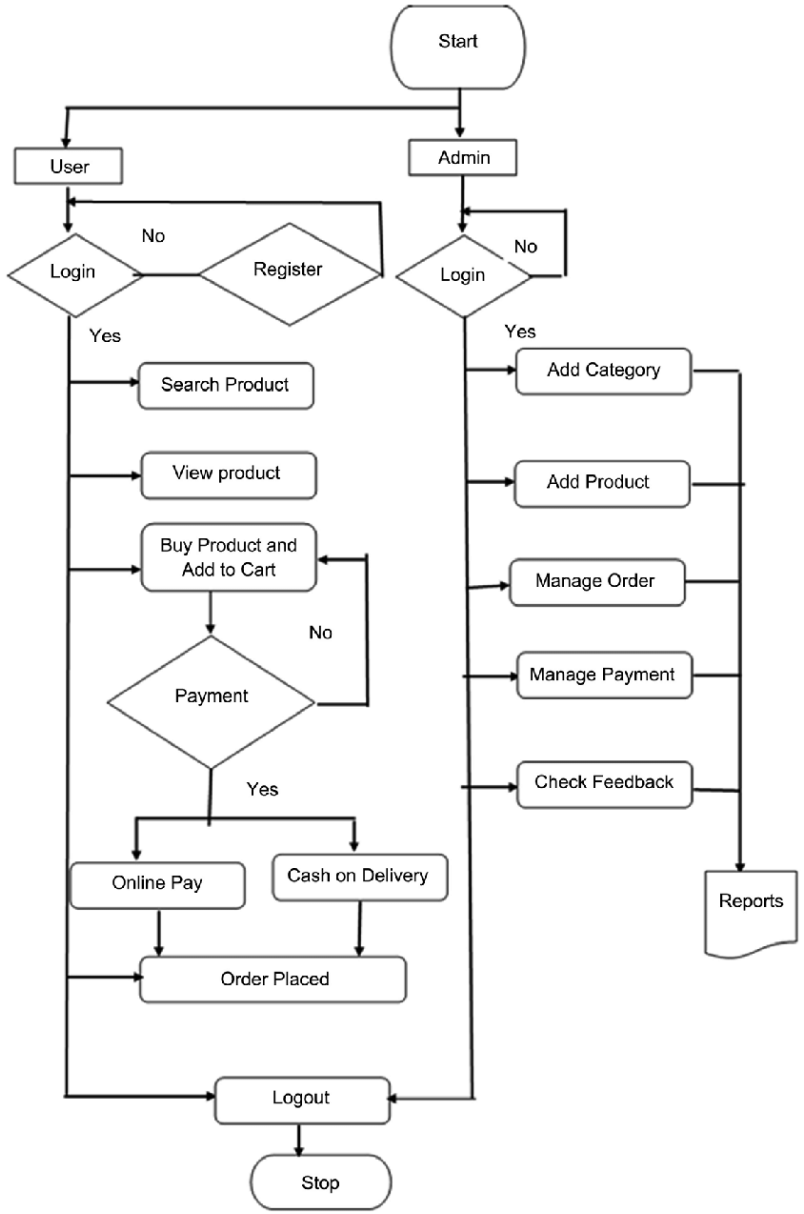
Assignment

- Assignment
  - Install the Python 3
    - Try pip install
    - Conda / Miniconda
    - Install Conda
    - Create an environment
  - Install the IDE that you like to use (i.e., Pycharm , Visual Studio Code)
    - Create a simple program
    - Try run, debug, breakpoint, check the variant value

- First lab from week 1

Outline

- Project
- **Control Flow**
  - **Conditional Control**
    - **if statement**
  - Loop Control
    - while
    - for
- Input/Output from console
- String format



https://www.researchgate.net/figure/Flow-chart-diagram-Online-e-commerce-shopping-web-app_fig1_347169962

Definition of Control Flow

- In computer science, control flow (or flow of control) is the order in which individual statements, instructions or function calls of an imperative program are executed or evaluated.

https://en.wikipedia.org/wiki/Computer_programming

Main Control Flow Structures

- **Sequential Control**
- **Conditional Control**
- **Loop Control**
- Function Call
- Exception Handling

Sequential Control

- The program executes instructions in the **order** they are written, from top to bottom.

```
first = 10
second = 20
print(first,second)
first,second = second,first
print(first,second)
```

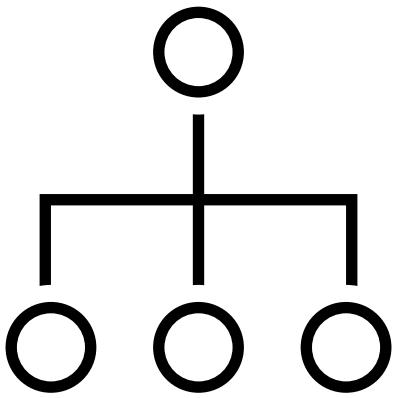
```
10 20
20 10
```

Conditional Control

- We would like the program to perform different steps depending on different conditions
- Decision-making ability

Conditional Control

- The program makes decisions based on the truth value of conditions, determining which path to take.
- Boolean variable type
  - True or False
- if-else statement
  - if, if-else, if-elif-else
- Boolean expression
  - An expression that evaluates to either True or False.



True and False

- *bool* is a subclass of *int*
  - True == 1
  - False == 0
  - True + True = 2

```
[1] print(True==1)
```

```
True
```

```
[2] print(False==0)
```

```
True
```

```
print(True+True)
```

```
2
```

Boolean Expression

- Logical Operator
  - and
  - or
  - not

a	b	a and b
True	True	True
True	False	False
False	True	False
False	False	False

a	b	a or b
True	True	True
True	False	True
False	True	True
False	False	False

Boolean Expression

- Logical Operator
  - and
  - or
  - not

a	not a
True	False
False	True

True and False

- Objects Evaluated as False
  - False itself
  - None
  - Zero of any numeric type (0, 0.0, 0j, etc.)
  - Empty containers ("", [], {}, set(), range(0))
- Objects Evaluated as True
  - All other values

```python
a = [1,2,3]
if a:
  print(f'{a} is True')
b = []
if b:
  print(f'{b} is True')
c = ''
if c:
  print(f'{c} is True')
d = 'testing'
if d:
  print(f'{d} is True')
```

```
[1, 2, 3] is True
testing is True
```

None*

- None is a singleton object in Python, of type NoneType.
- Represents “no value” or “nothing.”
- Default return value of a function with no return statement.
- Often used as a placeholder or to mark missing values.
- Evaluates to False in Boolean context.
- Always check with *is* None or *is not* None (not ==).

Identity Operator*

- Check whether two variables refer to the same object in memory.
- is
- is not
- Commonly used for None

```python
a = 10
b = 10
print(a is b)
print(id(a),id(b))
c = 10000
d = 10000
print(c is d)
print(id(c),id(d))
```

```
True
11642376 11642376
False
140080821566032 140080821560560
```

if

if expr:

statement(s)

```
drink = "Latte"
price = 0.0
if drink == "Latte":
    print("You chose Latte.")
    print("That will be $4.00.")
    price = 4.00
```

```
You chose Latte.
That will be $4.00.
```

Expressions in if Statements

```
if expr:
    statement(s)
```

1. Comparisons

- if x == 5:
- if a < b:
- Always return True or False
- *if x=5: → error*

2. Numbers

- if 0: → False
- if 10: → True

3. Containers

- if []: → False (empty list)
- if [1, 2, 3]: → True (non-empty)
- Works the same for strings, dicts, sets

4. None check

- Use explicit check:
- if x is None:
- …

0s

```
x=10
if x=9:
  print(x)
```

```
  File "/tmp/ipython-input-819998701.py", line 2
    if x=9:
       ^
SyntaxError: invalid syntax. Maybe you meant '==' or ':=' instead of '='?
```

Next steps: Explain error

if-else

if expr:

  statement(s)

else:

  statement(s)

```
drink = "Tea"
price = 0.0
if drink == "Latte":
    print("You chose Latte.")
    print("That will be $4.00.")
    price = 4.00
else:
    print("Sorry, we don't have that option.")
```

```
Sorry, we don't have that option.
```

if-elif-else

```
if expr1:
    statement(s)
elif expr2:
    statement(s)
else:
    statement(s)
```

if-elif-...-else

```
if expr1:
  statement(s)
elif expr2:
  statement(s)
elif expr3:
  statement(s)
else:
  statement(s)
```

```
drink = "Latte"
price = 0.0
if drink == "Americano":
    print("You chose Americano.")
    print("That will be $3.50.")
    price = 3.50
elif drink == "Latte":
    print("You chose Latte.")
    print("That will be $4.00.")
    price = 4.00
elif drink == "Tea":
    print("You chose Tea.")
    print("That will be $2.50.")
    price = 2.50
elif drink == "Hot Chocolate":
    print("You chose Hot Chocolate.")
    print("That will be $3.00.")
    price = 3.00
else:
    print("Sorry, we don't have that option.")
```

Details of if Statement

- The Colon (:)
  - The colon is required at the end of the if and else lines.
  - It indicates that a block of code will follow.
- Indentation:
  - Python uses indentation to define the scope of the code block.
  - Consistent indentation is crucial for the correct execution of the program.
  - Typically, four spaces or one tab is used for indentation.

Nested if Statement

- Be careful of indentations

```
drink = "Latte"
size = "Large"

if drink == "Latte":
    if size == "Large":
        print("You chose a Large Latte.")
        print("That will be $5.00.")
    else:
        print("You chose a Small Latte.")
    print("That will be $4.00.")
else:
    print("Sorry, we only have Latte available.")
```

Conditional Expression

- *value_if_true* if condition else value_if_false

```python
# Using a normal if-else statement
age = 20
if age >= 18:
    status = "adult"      # assign "adult" if condition is True
else:
    status = "minor"      # assign "minor" if condition is False
print(status)  # adult
```

```
adult
```

```python
# Using a conditional expression (ternary operator)
age = 20
status = "adult" if age >= 18 else "minor"   # one-line expression
print(status)  # adult
```

```
adult
```

Conditional Expression

- *value_if_true* if condition else value_if_false
- Is useful when you have a condition and you want to produce (return) a value based on that condition.

Programming Recommendations

- Use `is not` operator rather than `not ... is`. While both expressions are functionally identical, the former is more readable and preferred:

```
# Correct:
if foo is not None:
```

```
# Wrong:
if not foo is None:
```

- Don't compare boolean values to True or False using `==`:

```
# Correct:
if greeting:
```

```
# Wrong:
if greeting == True:
```

Worse:

```
# Wrong:
if greeting is True:
```

https://peps.python.org/pep-0008/

match-case Statement

```
match variable_name:
  case 'pattern 1' :
    statement1
  case 'pattern 2' :
    statement2
  ...
  case 'pattern n' :
    statement
  case _:
    print('default')
```

```
drink = "Latte"
price = 0.0
if drink == "Americano":
    print("You chose Americano.")
    print("That will be $3.50.")
    price = 3.50
elif drink == "Latte":
    print("You chose Latte.")
    print("That will be $4.00.")
    price = 4.00
elif drink == "Tea":
    print("You chose Tea.")
    print("That will be $2.50.")
    price = 2.50
elif drink == "Hot Chocolate":
    print("You chose Hot Chocolate.")
    print("That will be $3.00.")
    price = 3.00
else:
    print("Sorry, we don't have that option.")
```

In Python's match statement (introduced in Python 3.10), case _ is used as a **wildcard pattern**. It matches anything, similar to a "default" case in other languages like switch statements.

Review

- Boolean Expressions
- Logical Operators
  - and
  - or
  - not
- if elif elif elif else
- Conditional Expression
- match-case

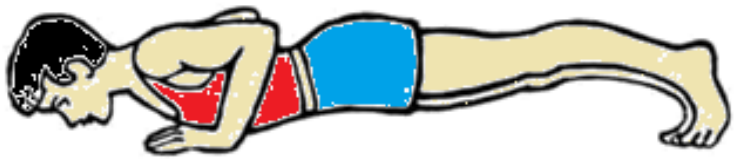
Outline

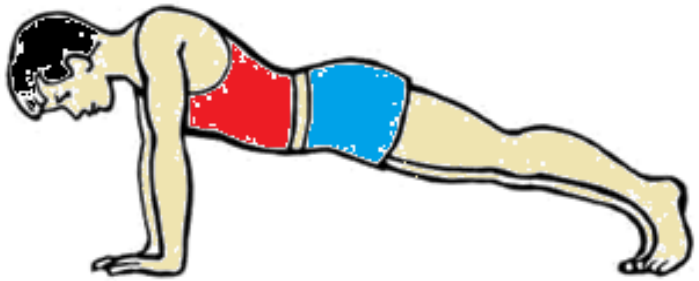
- Control Flow
  - Conditional Control
    - if statement
  - **Loop Control**
    - **while**
    - for
- Input/Output from console
- String format

https://www.researchgate.net/figure/Flow-chart-diagram-Online-e-commerce-shopping-web-app_fig1_347169962

Loop Control

- The program repeatedly executes a block of code as long as a specified condition is met.
- Structures: for, while
  - A person does 20 push-ups every day,
  - He stops once he has completed all 20 push-ups.





while Loop

- Sequential Control
  - Start
  - Push-up 1
  - Push-up 2
  - …
  - Push-up 20
  - End

- loop
  - Pseudocode:
    - pushup_count = 0
    - IF pushup_count < 20
      - Push-up
      - Increase pushup_count by 1
    - ELSE
      - Stop
    - IF pushup_count < 20
      - Push-up
      - Increase pushup_count by 1
    - ELSE
      - Stop
    - …

while Loop

The program repeatedly executes a block of code as long as a specified condition is met.

```
def PushDown():
    pass
def PushUp():
    pass
pushup_count = 0
while pushup_count < 20:
    print(pushup_count)
    PushDown()
    PushUp()
    pushup_count += 1
print("Completed 20 push-ups!")
print(pushup_count)
```

- loop
  - Pseudocode:
    - pushup_count = 0
    - IF pushup_count < 20
      - Push-up
      - Increase pushup_count by 1
    - ELSE
      - Stop
    - IF pushup_count < 20
      - Push-up
      - Increase pushup_count by 1
    - ELSE
      - Stop
    - …

pass

- Does nothing when executed.
- Used as a placeholder to keep code syntactically correct.

while Loop

```
while expression:
    statement1
    statement2
```

- Expression
  - Boolean Expression
  - Whether to stay within the loop
- Statements
  - Do something
  - Will change the result of the Expression

```python
def PushDown():
    pass
def PushUp():
    pass
pushup_count = 0
while pushup_count < 20:
    print(pushup_count)
    PushDown()
    PushUp()
    pushup_count += 1
print("Completed 20 push-ups!")
print(pushup_count)
```

Infinite Loops

- What happens if you forget pushup_coutnt += 1?
  - Dead Loop
  - The most common reason for a program ‘hanging’
  - To quit program execution: restart, Ctrl + C

- When Infinite Loop is useful?

  while True:
    - Continuous input loop → keep asking user until a quit command is given
    - Server or event listener → wait for connections or events indefinitely
    - Game loop → constantly update and render until game is over

The loop may not be executed

- If the condition is False to begin with

```python
def PushDown():
    pass
def PushUp():
    pass
pushup_count = 30
while pushup_count < 20:
    print(pushup_count)
    PushDown()
    PushUp()
    pushup_count += 1
print("Completed 20 push-ups!")
print(pushup_count)
```

```
Completed 20 push-ups!
30
```

Review

- Loop
  - The program
    - repeatedly executes a block of code
    - as long as a specified condition is met.

```
while expression:
    statement1
    statement2
```

- Infinite Loops

for Loop

```
def PushDown():
    pass
def PushUp():
    pass
for pushup_count in range(20):
    PushDown()
    PushUp()
    print(pushup_count)
print("Completed 20 push-ups!")
```

for iterating_var in sequence:

statements

- Define pushup_count inside for
- Sequence
  - range(20)

for Loop

- The for loop in Python provides the ability to loop over the items of any sequence, such as a list, tuple or a string.
- It performs the same action on each item of the sequence.

```python
def PushDown():
    pass
def PushUp():
    pass
for pushup_count in range(20):
    PushDown()
    PushUp()
    print(pushup_count)
print("Completed 20 push-ups!")
```

https://www.tutorialspoint.com/python/python_for_loops.htm

range()

- range(stop)
- range(start, stop)
- range(start, stop, step)
  - Start: starting num of the sequence
  - Stop: generate num up to, but **not including this num**
  - Step: difference between each num in the sequence

```python
x = range(10)
print(x)
print(type(x))
print(list(x))
```

```
range(0, 10)
<class 'range'>
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
```

```python
x = range(2,7)
print(type(x))
print(list(x))
```

```
<class 'range'>
[2, 3, 4, 5, 6]
```

```python
x = range(0,10,2)
print(list(x))
```

range()*

- Built-in function returning a range object, not a list
- Produces numbers on demand (lazy), not a full list
- Memory efficient: stores only start, stop, step
- Supports indexing & slicing
- Iterator and Generator **

```
[12] r = range(5)
     print(r)
```

```
range(0, 5)
```

```
[13] print(list(r))
```

```
[0, 1, 2, 3, 4]
```

```
[17] print(r[0])
```

```
0
```

```
print(r[1:3])
```

```
range(1, 3)
```

for Loop

- The for loop in Python provides the ability to loop over the items of any sequence, such as a list, tuple or a string.

```
def PushDown():
    pass
def PushUp():
    pass
for pushup_count in range(20):
    PushDown()
    PushUp()
    print(pushup_count)
print("Completed 20 push-ups!")
```

Compare for and while

```
while boolean_expression:
    Statements
```

- Condition met or not

```
for iterating_var in sequence:
    Statements
```

- Go through the sequence
- List, tuple, string, range
- Iterable:
  - Whenever we say an iterable in Python, it means a sequence data type (for example, a list).

for Loop

- The for loop in Python provides the ability to loop over the items of **any** sequence, such as a list, tuple or a string.
- String as a sequence

```
name = 'taylor swift'
for char in name:
    print(char)
```

```
t
a
y
l
o
r

s
w
i
f
t
```

Dead loop

- loop over the items of a sequence
- How can we write a dead loop in 'for'?

```
[23] for i in range(4):
         i-=1
         print(i)
```

```
-1
0
1
2
```

Dead loop

```python
ll = [0,1,2,3]
for a in ll:
  ll.append(a+1)
  print(ll)
```

```
[0, 1, 2, 3, 1]
[0, 1, 2, 3, 1, 2]
[0, 1, 2, 3, 1, 2, 3]
[0, 1, 2, 3, 1, 2, 3, 4]
[0, 1, 2, 3, 1, 2, 3, 4, 2]
[0, 1, 2, 3, 1, 2, 3, 4, 2, 3]
[0, 1, 2, 3, 1, 2, 3, 4, 2, 3, 4]
[0, 1, 2, 3, 1, 2, 3, 4, 2, 3, 4, 5]
[0, 1, 2, 3, 1, 2, 3, 4, 2, 3, 4, 5, 3]
[0, 1, 2, 3, 1, 2, 3, 4, 2, 3, 4, 5, 3, 4]
[0, 1, 2, 3, 1, 2, 3, 4, 2, 3, 4, 5, 3, 4, 5]
[0, 1, 2, 3, 1, 2, 3, 4, 2, 3, 4, 5, 3, 4, 5, 6]
[0, 1, 2, 3, 1, 2, 3, 4, 2, 3, 4, 5, 3, 4, 5, 6, 4]
[0, 1, 2, 3, 1, 2, 3, 4, 2, 3, 4, 5, 3, 4, 5, 6, 4, 5]
[0, 1, 2, 3, 1, 2, 3, 4, 2, 3, 4, 5, 3, 4, 5, 6, 4, 5, 6]
[0, 1, 2, 3, 1, 2, 3, 4, 2, 3, 4, 5, 3, 4, 5, 6, 4, 5, 6, 7]
[0, 1, 2, 3, 1, 2, 3, 4, 2, 3, 4, 5, 3, 4, 5, 6, 4, 5, 6, 7, 5]
[0, 1, 2, 3, 1, 2, 3, 4, 2, 3, 4, 5, 3, 4, 5, 6, 4, 5, 6, 7, 5, 6]
[0, 1, 2, 3, 1, 2, 3, 4, 2, 3, 4, 5, 3, 4, 5, 6, 4, 5, 6, 7, 5, 6, 7]
[0, 1, 2, 3, 1, 2, 3, 4, 2, 3, 4, 5, 3, 4, 5, 6, 4, 5, 6, 7, 5, 6, 7, 8]
[0, 1, 2, 3, 1, 2, 3, 4, 2, 3, 4, 5, 3, 4, 5, 6, 4, 5, 6, 7, 5, 6, 7, 8, 6]
[0, 1, 2, 3, 1, 2, 3, 4, 2, 3, 4, 5, 3, 4, 5, 6, 4, 5, 6, 7, 5, 6, 7, 8, 6, 7]
[0, 1, 2, 3, 1, 2, 3, 4, 2, 3, 4, 5, 3, 4, 5, 6, 4, 5, 6, 7, 5, 6, 7, 8, 6, 7, 8]
[0, 1, 2, 3, 1, 2, 3, 4, 2, 3, 4, 5, 3, 4, 5, 6, 4, 5, 6, 7, 5, 6, 7, 8, 6, 7, 8, 9]
[0, 1, 2, 3, 1, 2, 3, 4, 2, 3, 4, 5, 3, 4, 5, 6, 4, 5, 6, 7, 5, 6, 7, 8, 6, 7, 8, 9, 7]
[0, 1, 2, 3, 1, 2, 3, 4, 2, 3, 4, 5, 3, 4, 5, 6, 4, 5, 6, 7, 5, 6, 7, 8, 6, 7, 8, 9, 7, 8]
[0, 1, 2, 3, 1, 2, 3, 4, 2, 3, 4, 5, 3, 4, 5, 6, 4, 5, 6, 7, 5, 6, 7, 8, 6, 7, 8, 9, 7, 8, 9]
```

Break Statement

- Stop the loop when some condition is satisfied
- Can be used in both Python while and for loops

```python
for i in range(10):
    print(i)
    if i == 5:
        break
```

```
0
1
2
3
4
5
```

Continue Statement

- Skips the remaining statements in the current loop and starts the next iteration.
- Can be used in both Python while and for loops
  - Example:
  - I don't do exercise on Sunday

```python
def PushDown():
    pass
def PushUp():
    pass
pushup_count = 0
day = 0
while pushup_count < 20:
    day += 1
    if day%7==0:
      print(f'day {day}, pushup_count {pushup_count}')
      continue
    PushDown()
    PushUp()
    pushup_count += 1
    print(f'day {day}, pushup_count {pushup_count}')
print("Completed 20 push-ups!")
print(pushup_count)
```

for ... else ...

```
students = ["b", "c", "d"]
found = False

for s in students:
    if s == "a":
        print("Set score = 0")
        found = True
        break

if not found:
    print("No such student")
```

```
students = ["b", "c", "d"]

for s in students:
    if s == "a":
        print("Set score = 0")
        break
else:
    print("No such student")
```

for … else …

- It's not that the `else` is skipped whenever the `for` loop runs.
- The `else` is skipped only if the loop is terminated by a `break`.
  - Otherwise, the `else` will still run.

```
students = ['b','c','d']
for s in students:
  if s=='a':
    print('Set score = 0')
    break
else:
  print('No such student')
  print('Everyone is happy')
```

```
No such student
Everyone is happy
```

```
[20] students = ['a','b','c','d']
     for s in students:
       if s=='a':
         print('Set score = 0')
         break
     else:
       print('No such student')
       print('Everyone is happy')
```

```
Set score = 0
```

Review

- Loop over the items of a sequence

```
for iterating_var in sequence:
    Statements
```

- Break statement
- Continue statement

Input and Output

- Program needs users to provide some data
- Two main different input manners
  - Input from the Keyboard
  - Input from files

Input from the keyboard

```
mark@linux-desktop: /tmp/tutorial
File  Edit  View  Search  Terminal  Help
mark@linux-desktop:~$ mkdir /tmp/tutorial
mark@linux-desktop:~$ cd /tmp/tutorial
mark@linux-desktop:/tmp/tutorial$ mkdir dir1 dir2 dir3
mark@linux-desktop:/tmp/tutorial$ mkdir
mkdir: missing operand
Try 'mkdir --help' for more information.
mark@linux-desktop:/tmp/tutorial$ cd /etc ~/Desktop
bash: cd: too many arguments
mark@linux-desktop:/tmp/tutorial$ ls
dir1  dir2  dir3
mark@linux-desktop:/tmp/tutorial$
```

https://ubuntu.com/tutorials/command-line-for-beginners#4-creating-folders-and-files

input()

- input ()
  - Read one line from standard input
  - Return it as a string
  - We can give a prompt text
    - which will appear before the cursor when you run the code.

```
a = input('tell me are you bored: ')
print(a)
print('tell me are you bored: ')
b = input()
print(b)
```

```
tell me are you bored: no of course
no of course
tell me are you bored:
yes sure
yes sure
```

Type casting

- input ()
  - Read one line from standard input
  - Return it as a string
  - You need to convert the data type if necessary

```
distance_pre = input('how many meters did you walk yesterday? ')
distance_today = input('how many meters did you walk today? ')
total_distance = distance_pre+distance_today
print('oh i know! you walked a total of ')
print(total_distance)
```

```
how many meters did you walk yesterday? 100
how many meters did you walk today? 200
oh i know! you walked a total of
100200
```

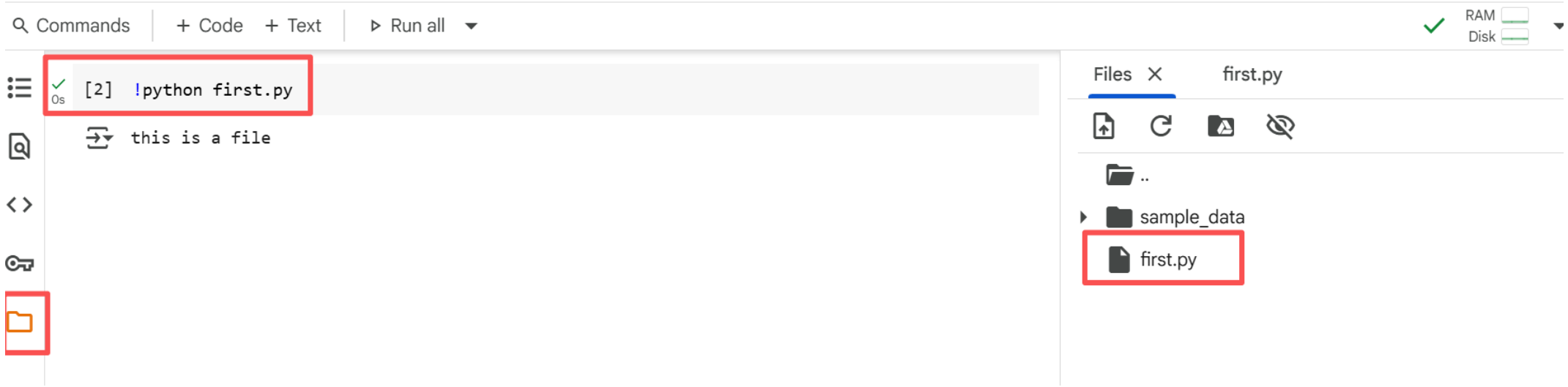
```
distance_pre = input('how many meters did you walk yesterday? ')
distance_today = input('how many meters did you walk today? ')
total_distance = int(distance_pre)+int(distance_today)
print('oh i know! you walked a total of ')
print(total_distance)
```

```
how many meters did you walk yesterday? 100
how many meters did you walk today? 200
oh i know! you walked a total of
300
```

Writing a Python File in Google Colab

- Click the Files tab on the left sidebar.
- On the right, click Refresh if needed.
- Right Click, Click New File, and save it as first.py.
- Edit the file content:
  - print("Hello, Python!")
- Save the file.
- Run it with:
  - !python first.py ⚠️
- Note: Don’t forget the exclamation mark (!) in front of python.

Writing a Python File in Google Colab



Introduction to sys.argv

- sys.argv is a list in Python
- contains the command-line arguments passed to the script
- sys.argv[0] is the name of the script
- sys.argv[1] to sys.argv[n] are the additional arguments passed

Introduction to sys.argv

Commands + Code + Text Run all RAM Disk

```
!python first.py 1 2 3
```

```
['first.py', '1', '2', '3']
<class 'list'>
```

Files first.py

```python
import sys
print(sys.argv)
print(type(sys.argv))
```

__name__ and __main__*

Commands + Code + Text Run all RAM Disk

```
!python first.py 1 2 3
```

```
['first.py', '1', '2', '3']
<class 'list'>
```

Files first.py

```python
import sys
def show_args():
  print(sys.argv)
  print(type(sys.argv))

if __name__=='__main__':
  show_args()
```

__name__ and __main__*

- __name__
  - A special variable assigned by the Python Interpreter
  - Tells us how the file is being used
  - "__main__" is one possible value of __name__
    - When the file is executed directly
    - The entry point of a Python program
  - otherwise the value of __name__ is the module's name
    - "first"

Vale of __name__*

hello.py

python Copy code

```python
print("In hello.py, __name__ =", __name__)
```

test.py

python Copy code

```python
import hello
```

```
[7] !python hello.py
In hello.py, __name__ = __main__

!python test.py
In hello.py, __name__ = hello
```

Output to the screen

- Print one variable's value
  - print ("something or null" + variable_name)
- Print one string variable (e.g. name)
  - print ("this is one string" + name)
- Print one int variable (e.g. age)
  - print (age)
  - print ("Your age is: " + str(age))
- Print one float variable (e.g. salary)
  - print (salary)
  - print ("Your age is: " + str(salary))

Neat and organized print format

```python
tom_salary = 12345.25
jim_salary = 123.5
mike_salary = 12345678.5
jack_salary = 1.23

tom_name = 'Tome Grace Liu'
jim_name = 'Jim'
mike_name = 'Mike Raghu Raghu'
jack_name = 'Jack Li'

print('name    salary')
print(tom_name+'    '+str(tom_salary))
print(jim_name+'    '+str(jim_salary))
print(mike_name+'    '+str(mike_salary))
print(jack_name+'    '+str(jack_salary))
```

```
name    salary
Tome Grace Liu    12345.25
Jim    123.5
Mike Raghu Raghu    12345678.5
Jack Li    1.23
```

Looks better?

```
name     salary
Tome Grace Liu     12345.25
Jim     123.5
Mike Raghu Raghu     12345678.5
Jack Li     1.23
```

```
name                salary
Tome Grace Liu      12345.25
Jim                 123.50
Mike Raghu Raghu    12345678.50
Jack Li             1.23
```

```
                name          salary
   Tome Grace Liu          12345.25
              Jim            123.50
Mike Raghu Raghu       12345678.50
         Jack Li              1.23
```

Alignment problem
How to align them?

Data Formatting: f-string

- Print variables

```
tom_salary = 12345.25
jim_salary = 123.5
mike_salary = 12345678.5
jack_salary = 1.23

tom_name = 'Tome Grace Liu'
jim_name = 'Jim'
mike_name = 'Mike Raghu Raghu'
jack_name = 'Jack Li'

print('name    salary')
print(tom_name+'    '+str(tom_salary))
print(f'{tom_name}    {tom_salary}')
print(jim_name+'    '+str(jim_salary))
print(f'{jim_name}    {jim_salary}')
```

```
name    salary
Tome Grace Liu    12345.25
Tome Grace Liu    12345.25
Jim    123.5
Jim    123.5
```

f-string

- Print expressions

```
price = 10
quantity = 3
fstring = f'Price: {price}    Quantity : {quantity}    Total : {price*quantity}'
print (fstring)
```

```
Price: 10    Quantity : 3    Total : 30
```

f-string

- Alignment
- {variable:>width}
- > right-aligned
- Width: the total number of characters

```python
tom_salary = 12345.25
jim_salary = 123.5
mike_salary = 12345678.5
jack_salary = 1.23

tom_name = 'Tome Grace Liu'
jim_name = 'Jim'
mike_name = 'Mike Raghu Raghu'
jack_name = 'Jack Li'

print('name    salary')
print(f'{tom_name}    {tom_salary}')
print(f'{jim_name}    {jim_salary}')
print('')
print('12345678901234567890    1234567890')
name='name'
salary='salary'
print(f'{name:>20}    {salary:>10}')
print(f'{tom_name:>20}    {tom_salary:>10}')
print(f'{jim_name:>20}    {jim_salary:>10}')
```

```
name    salary
Tome Grace Liu    12345.25
Jim    123.5

12345678901234567890    1234567890
                name        salary
      Tome Grace Liu      12345.25
                 Jim         123.5
```

f-string

- Alignment
- {variable:>width}
- {variable:<width}
- {variable:^width}
- > right-aligned
- < left-aligned
- ^ caret sign, center-aligned
- Width: the total number of characters

```python
tom_salary = 12345.25
jim_salary = 123.5
mike_salary = 12345678.5
jack_salary = 1.23

tom_name = 'Tome Grace Liu'
jim_name = 'Jim'
mike_name = 'Mike Raghu Raghu'
jack_name = 'Jack Li'

print('name    salary')
print(f'{tom_name}    {tom_salary}')
print(f'{jim_name}    {jim_salary}')
print('')
print('12345678901234567890    1234567890')
name='name'
salary='salary'
print(f'{name:>20}    {salary:>10}')
print(f'{tom_name:>20}    {tom_salary:>10}')
print(f'{jim_name:>20}    {jim_salary:>10}')
```

```
name    salary
Tome Grace Liu    12345.25
Jim    123.5

12345678901234567890    1234567890
                name        salary
      Tome Grace Liu      12345.25
                 Jim         123.5
```

f-string

- Alignment
- {variable:>width}
- {variable:<width}
- {variable:^width}
- > right-aligned
- < left-aligned
- ^ caret sign, center-aligned
- Width: the total number of characters

```python
tom_salary = 12345.25
jim_salary = 123.5
mike_salary = 12345678.5
jack_salary = 1.23

tom_name = 'Tome Grace Liu'
jim_name = 'Jim'
mike_name = 'Mike Raghu Raghu'
jack_name = 'Jack Li'

print('name    salary')
print(f'{tom_name}    {tom_salary}')
print(f'{jim_name}    {jim_salary}')
print('')
print('12345678901234567890    1234567890')
name='name'
salary='salary'
print(f'{name:<20}    {salary:<10}')
print(f'{tom_name:<20}    {tom_salary:<10}')
print(f'{jim_name:<20}    {jim_salary:<10}')
```

```
name    salary
Tome Grace Liu    12345.25
Jim    123.5

12345678901234567890    1234567890
name                    salary
Tome Grace Liu          12345.25
Jim                     123.5
```

f-string

- Precision handling of floats
- {variable:>[width].[precision]f}
- Width: total number of digits
- Precision: decimal digits
- Do not forget f
- What will happen if we lose f?

```
salary = 543.255
print('your salary is 1234567890, right?')
print(f'your salary is {salary:<10.2f}, right?')
print(f'your salary is {salary:>10.2f}, right?')
print(f'your salary is {salary:>10.4f}, right?')
salary = 3.1415
print(f'your salary is {salary:>10.3f}, right?')
```

```
your salary is 1234567890, right?
your salary is 543.25    , right?
your salary is     543.25, right?
your salary is   543.2550, right?
your salary is      3.142, right?
```

Data Formatting

- Format strings
  - %<width>**s**

```
name = 'Mike'
print('your name is 1234567, right?')
print('your name is %7s, right?'%name)
```

```
your name is 1234567, right?
your name is    Mike, right?
```

- Format integers
  - %<width>**d**

```
age = 80
print('your age is 12345, right?')
print('your age is %5d, right?'%age)
```

```
your age is 12345, right?
your age is    80, right?
```

Data Formatting

- Format a floating point number
  - %<width>.<precision>f
    - Percent sign followed by width, dot, precision, and f
  - Width: total length of the number, including the decimal point
  - Precision: number of decimal places
  - If width < real_length, Print real_length

```
salary = 543.21
print('your salary is 12345678, right?')
print('your salary is %8.3f, right?'%salary)
```

```
your salary is 12345678, right?
your salary is  543.210, right?
```

```
salary = 543.28
print('your salary is 12345678, right?')
print('your salary is %8.1f, right?'%salary)
```

```
your salary is 12345678, right?
your salary is    543.3, right?
```

```
salary = 543.26
print('your salary is 12345678, right?')
print('your salary is %2.2f, right?'%salary)
```

```
your salary is 12345678, right?
your salary is 543.26, right?
```

Further Reading

round(*number, ndigits=None*)

Return *number* rounded to *ndigits* precision after the decimal point. If *ndigits* is omitted or is `None`, it returns the nearest integer to its input.

For the built-in types supporting `round()`, values are rounded to the closest multiple of 10 to the power minus *ndigits*; if two multiples are equally close, rounding is done toward the even choice (so, for example, both `round(0.5)` and `round(-0.5)` are `0`, and `round(1.5)` is `2`). Any integer value is valid for *ndigits* (positive, zero, or negative). The return value is an integer if *ndigits* is omitted or `None`. Otherwise, the return value has the same type as *number*.

For a general Python object `number`, `round` delegates to `number.__round__`.

> **Note:** The behavior of `round()` for floats can be surprising: for example, `round(2.675, 2)` gives `2.67` instead of the expected `2.68`. This is not a bug: it's a result of the fact that most decimal fractions can't be represented exactly as a float. See Floating-Point Arithmetic: Issues and Limitations for more information.

```python
a = 2.675
print(f'{a=}')
b = f'{a:.2f}'
print(f'round a {b}')
a = 2.6751
print(f'{a=}')
b = f'{a:.2f}'
print(f'round a {b}')
print(2.675>2.6749999999999999)
```

```
a=2.675
round a 2.67
a=2.6751
round a 2.68
False
```

How is a decimal represented in a computer?

https://docs.python.org/3/library/functions.html#round

Review

- Input from screen
- Output to screen
- f-string
  - Alignment
  - String width control
  - Decimal precision control

Git

- Git and GitHub are different
- Version Control
  - Forget what has been changed two weeks ago
  - How to combine many people's work
- Go to https://git-scm.com/downloads

Git

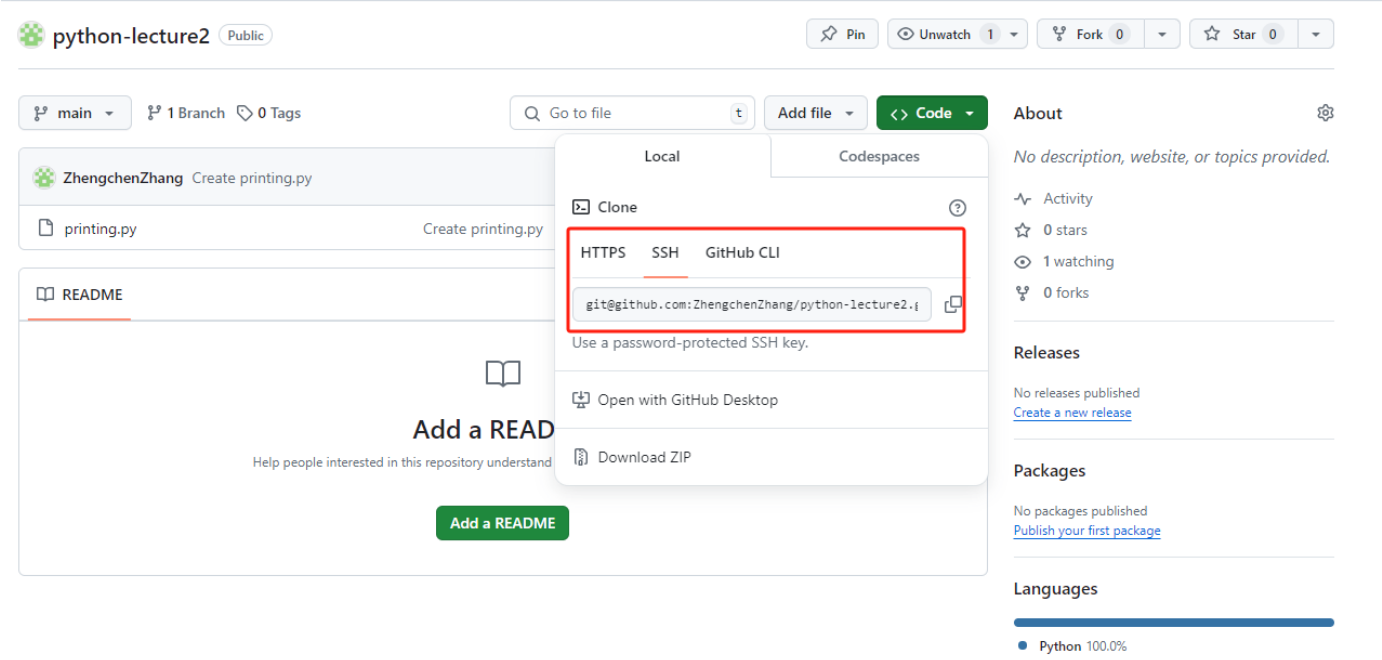
1. Go to the command line
   - Windows + R, input *cmd*
2. git config --global user.name "Mona Lisa"
   - If your name has a whitespace, then the double quotes are necessary
3. git config --global user.email "your.email@example.com"
   - Not for logging into any website
   - To record who submit the codes

Git

1. Go to a folder
   - cd d:\
   - cd ~
2. git init project1
   - Will create a folder named project1
3. cd project1
4. Make a txt file: file1.txt
   - Write something and save and close
5. git add file1.txt
6. git commit –m "my first commit"

Git

- Create a public repository on GitHub
- git clone <repository_url>
- https://github.com/ZhengchenZhang/python-lecture2.git
- git branch
- git checkout –b main
- Modify the files
- git add *.py
- git commit –m "my homework"
- git remote add origin <repository_url>
- git push origin main

Secure Shell Protocol (SSH)

1. ssh-keygen -t rsa -b 4096 -C your_email@example.com
2. Type *enter* many times
3. C:\users\<user_name>\.ssh\id_rsa.pub
4. Open it using a text editor, copy the content
5. Go to https://github.com/settings/keys
6. Click *New SSH key*
7. Give any name, and paste the content
8. Get the ssh address of your repository
9. Repeat the comments in the last page